AI SALES AGENT

Architecture overview

The modular monolith, its modules, the conversation rules, and the decisions that constrain the design.

Phase 1 MVP · 21 October 2026 · Prepared from the canonical documents in docs/

Architectural style

The system is a modular monolith. One NestJS application is deployed, and the backend and the worker are the same image with different commands. Domain modules talk through application services and domain events. A module does not read or write another module's tables. That rule is a review failure, not a performance suggestion.

Microservices were rejected for the MVP because the team is small, the purchase handoff is one unit of work rather than a distributed saga, and tenancy is already expressed as `business_id` on a shared PostgreSQL database. A single unstructured Nest module was also rejected: the channel adapter, the inbox, and the orchestrator must not leak into each other. A module is extracted later only for an independent scale need, an independent failure domain, or an independent team. The documented likely order starts with Integrations.

What the product is doing

AI Sales Agent is an AI sales employee plus a human sales desk for merchants who sell in Instagram and Messenger. It is not a generic chatbot, a storefront, a payment agent, a CRM, or a helpdesk. The AI handles most of the conversation. A person closes the sale. In the MVP there is no payment gateway.

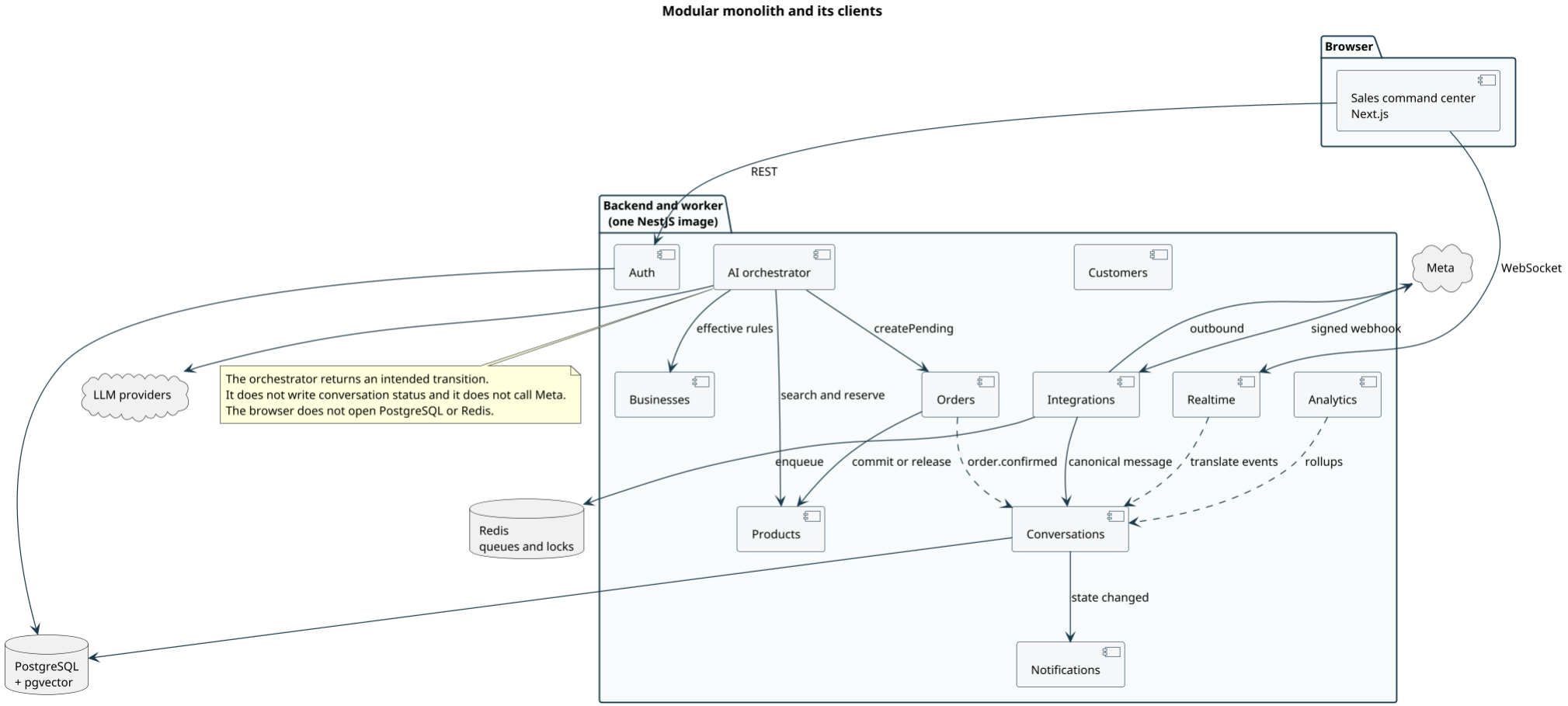
Three properties are load-bearing. The webhook persists the event and returns 200 before any model call. Commerce facts enter a reply only from a tool result or a stored business rule in the same turn. Conversation status is changed only by the conversation service. The model may request a transition. It does not write the status.

Containers

Container	Responsibility
frontend	Next.js sales command center. No business rules, no database, no Meta calls.
backend	REST under /api/v1, Meta webhooks, and the Socket.IO gateway. Persists inbound work and enqueues it.
worker	BullMQ processors: AI turns, outbound delivery, embeddings, notifications, and rollups.
PostgreSQL 16 + pgvector	System of record, including embeddings.
Redis 7	Queues, conversation locks, rate limits, and the Socket.IO adapter. Not a source of truth.
S3	Images, CSV originals, knowledge files, and conversation media. The database stores keys and URLs, not blobs.

Component view

Major components inside the one NestJS image, and the systems they are allowed to call. Source: `submission/sources/diagram-source-files/13-component.puml`.



Domain modules

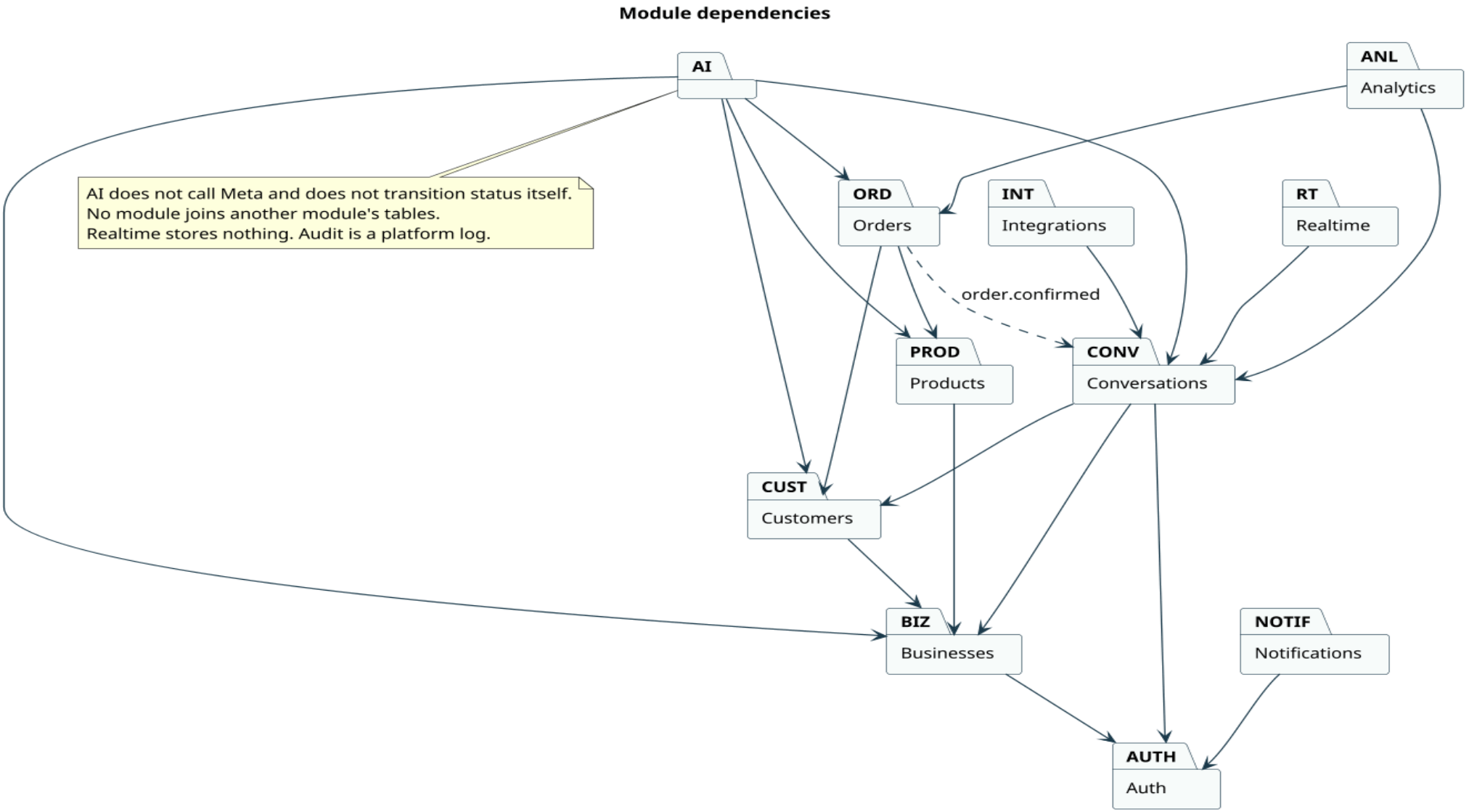
Module	Owns	Must not
Auth	Users, passwords, refresh tokens, OAuth links	Know about conversations or Meta
Businesses	Workspace, members, roles, rules, flags, plans, onboarding	Call Meta or a model

Module	Owns	Must not
Products	Catalog, variants, images, stock, reservations, knowledge, embeddings	Send messages or change conversation status
Customers	Profiles, identities, tags, notes, lifecycle	Own conversations or orders
Conversations	Threads, messages, assignments, summaries, lead score, status transitions	Call Meta or a model directly
AI	Agent, prompts, orchestrator, traces, usage, feedback	Persist orders or messages except through their owners
Orders	Orders, lines, confirm, cancel	Call Meta, or change stock except through Products
Integrations	Channel accounts, tokens, adapters, raw events, outbound delivery	Import the orchestrator or inbox internals
Analytics	Daily KPI rollups and usage meters	Sit on the inbound hot path
Notifications	In-app records and fan-out	Decide that a handoff is required
Realtime	Socket.IO rooms and client events. No tables.	Be a source of truth

Audit logs are a platform concern called by other modules. They are not a twelfth product module. Realtime joins rooms on the server. A client cannot subscribe to an arbitrary room. Workers reject a job that has no business id.

Module dependencies

Dependencies point at the owning module's service. The dashed arrow is an event, not a shared table. Source: `submission/sources/diagram-source-files/16-package-modules.puml`.



Tenancy and security boundary

Every tenant table carries `business_id`. Isolation is enforced in the application on every query. PostgreSQL row-level security is off in the MVP by decision, not by omission. A missing tenant filter is a severity-1 defect. Users belong to businesses through memberships. The user table itself is global. Roles are owner, admin, and employee. The merchant interface may label employee as Agent. There is no second owner and no custom role in this phase.

Provider tokens and API keys live in Secrets Manager. The channel row stores a reference. Webhooks are accepted only after the signature check. Customer phone numbers and addresses are redacted from logs and from prompts where the task does not need them. Authorisation is checked on the server. Hiding a button is not a control.

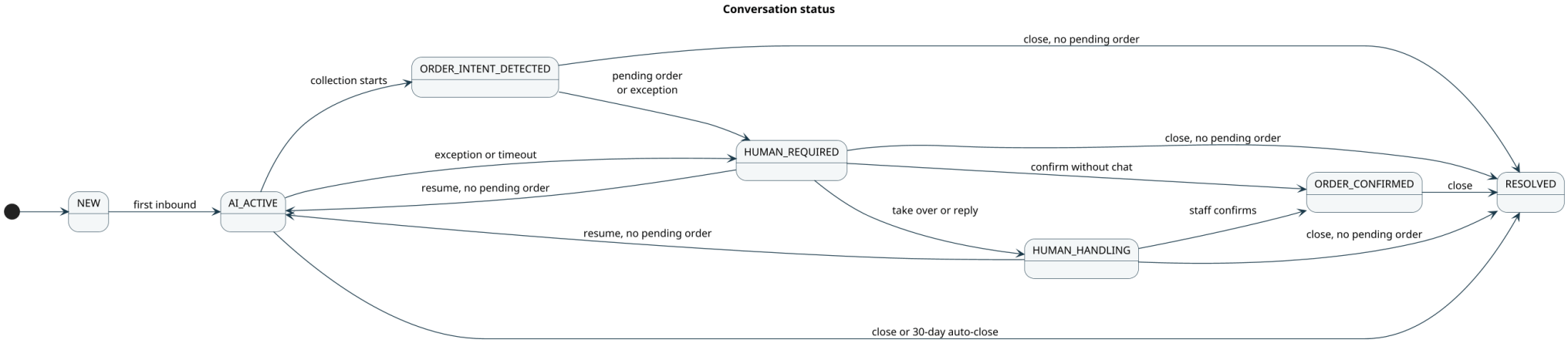
Conversation behaviour

A conversation has one status. Every change is appended to a transition log. Entering human-required stores the escalation reason, the priority, and a staff briefing before the last autonomous reply, and it sets `ai_enabled` to false. Fifteen product triggers ship in the MVP, from purchase handoff to integration failure. A model timeout is an escalation reason on the same exception path. It is not a sixteenth product trigger. When two triggers match, the higher priority wins and the other is kept on the briefing.

Reply mode is `auto_send`, `approval_required`, or `draft_only`. A new business starts in approval required. Draft only is Should. An approved or edited AI draft counts as AI-sent, with the edit recorded. Suggestions are generated only after an explicit staff request. They are never delivered by themselves.

Conversation state machine

The only legal statuses. Unknown pairs return HTTP 409. Resume and close are refused while a pending order exists. Source: `submission/sources/diagram-source-files/15-state-conversation.puml`.



Lead score and the inbox

Lead score is a deterministic 0–100 triage signal. It is not a purchase probability. Temperature bands are hot, warm, and cold. The inbox does not sort by recency. Human-needed threads come first, by handoff priority and then by wait. AI-handled threads follow by temperature and score. Notifications fire for a hot lead, an order request, a handoff, or an escalation. A correctly working agent produces a quiet desk.

Technology decisions

Concern	Choice
API	TypeScript, NestJS, REST under /api/v1. Fastify is the HTTP adapter (ADR-025).
Data	PostgreSQL 16, pgvector, Prisma. Money is numeric(12,3), never a float. Identifiers are UUID v4.
Jobs and realtime	Redis, BullMQ, Socket.IO on the backend task.
Operator UI	Next.js on ECS, not on Vercel. French, Arabic RTL, and English.
Customer reply model	gemini-3.5-flash, then gpt-5.6-terra, then claude-sonnet-5.
Classifier and embeddings	gpt-5.6-luna and text-embedding-3-small.

Concern	Choice
Channels	Instagram Messaging and Messenger. WhatsApp is a stub behind a flag until Phase 2. Unofficial WhatsApp automation is prohibited.

The decision index is ADR-001 through ADR-025 in the project decision log. This overview does not reopen those decisions. Deployment nodes, security groups, and the release pipeline are in the deployment document.